

Phase-One LHeC

Contact person:

Krzysztof Piotrkowski
AGH University, Kraków, Poland
E-mail: piotrkowski@agh.edu.pl
(Dated: March 31, 2025)

On behalf of Kevin D. J. André, Laurent Forthomme, Bernhard Holzer, and Krzysztof Piotrkowski.

Abstract

In *White Paper* on the Large Hadron electron Collider (LHeC), submitted to the ESPP Update [1], its commissioning is proposed several years after the completion of the HL-LHC programme in 2041. Here an alternative staged approach is considered in which a phase-one LHeC is using a 20 GeV single-pass Energy Recovery Linac (ERL), to be commissioned at P2 after LS4 allowing for unique studies of the electron-hadron interactions already during Run5. In this case, the project financial stress will be distributed over more years, and the experience acquired during the first phase of the LHeC will allow for a more robust design of the final stand-alone LHeC and possibly also allow its earlier startup. Moreover, such a phase-one LHeC is expected to provide important and timely scientific feedback to the hadron-hadron experiments at the HL-LHC already two years after commissioning, especially in the domain of the parton distribution functions. Conceptual studies indicate high luminosity performance [2], and this in turn guarantees excellent original scientific output, including, for example, unique research in the physics of the Higgs boson, top quark and $\gamma\gamma$ interactions at the electron-proton centre-of-mass energy of 0.75 TeV, as well as interesting comparative studies of eA and AA interactions, since the proposed design of the interaction region allows for both types of collisions at IP2 [3]. Last but not least, the new detector at the P2 site can be designed by extension of the ALICE 3 proposal, which offers strong synergies and benefits in several areas.

Phase-One LHeC as an essential complement to the present HL-LHC programme – the first stage of Large Hadron electron Collider (LHeC), as sketched in Figure 1, offers a compelling and optimal complement to the HL-LHC program. An Energy Recovery Linac (ERL) based accelerator delivers 20 GeV high-intensity electron beams to collide at IP2 with the proton or ion beams at centre-of-mass energies close to the TeV per nucleon, and luminosities well above $10^{33} \text{ cm}^{-2} \text{ s}^{-1}$.

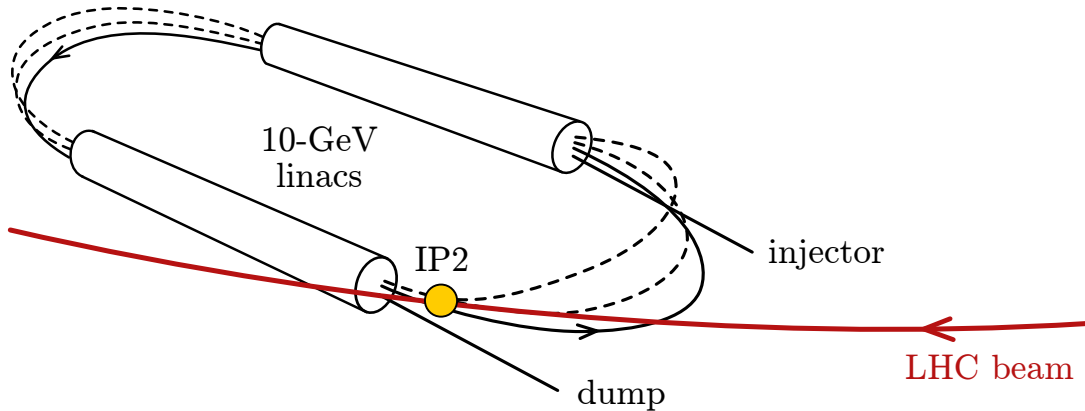


FIG. 1. Schematics of the single-pass LHeC layout with the optional return arcs 3 – 6 (dashed lines).

Such a staged realisation of the LHeC project provides a large benefit with respect to the non-staged proposal not only in reducing financial strains but first of all in the availability of concurrent electron-hadron collisions already during Run5 of the HL-LHC. Consequently, apart from the possibility of performing a wide range of unique and impactful studies almost 10 years earlier, the results of such a new electron-hadron experiment could also be used to timely improve measurements in other (hadron-hadron) experiments at the HL-LHC.

As discussed in the LHeC CDR [4], a relatively low integrated luminosity of 50 fb^{-1} is sufficient to greatly improve the constraints on the parton density functions (PDFs), particularly at low x . Therefore, already after two years of running the phase-one LHeC it will be possible to provide valuable feedback to hadron-hadron experiments at the HL-LHC, what will result in significant improvements in a number of precision measurements [1]. Moreover, with entering into a new kinematic region of very low x the novel QCD phenomena are expected to be observed – new parton dynamics and nonlinear parton density effects. It should also be stressed that the phase-one LHeC provides an important opportunity to perform eA and AA measurements using the same detector. Finally, with a total integrated luminosity of 250 fb^{-1} expected during Run5 for the ep collision mode, a number of relevant measurements can be made particularly in the domains of the Higgs boson, top quark, and $\gamma\gamma$ interactions.

The achievement of high-luminosity ep collisions with the LHeC is enabled by the Energy Recovery Linac (ERL) accelerator technology. This innovative technology addresses the energy sustainability of modern accelerators by recirculating the beam power rather than the particles in the beam.

The physics potential of the phase-one LHeC extending the HL-LHC programme – In the first phase of LHeC, the centre-of-mass energy for electron-proton collisions $\sqrt{s_{ep}} = \sqrt{4E_e E_p} = 0.75 \text{ TeV}$, in place of 1.2 TeV for the 50 GeV electron beam at the final LHeC. This means that the kinematical reach on the $x - Q^2$ plane will already be greatly extended with respect to the data taken at HERA at $\sqrt{s} = 0.3 \text{ TeV}$, as $Q_{\text{max}}^2 \simeq s_{ep}$ and $x_{\text{min}} \propto 1/s_{ep}$.

The ep collisions from low- x to high- x values enable the precise study of QCD physics. Even with only 50 fb^{-1} of phase-one LHeC data, the uncertainty on α_s can be reduced to about 0.2% and the running of α_s can be measured with few per mille-level accuracy. The improved measurements of PDFs and of α_s will greatly reduce the dominant source of uncertainty in the $gg \rightarrow H$ cross section at the HL-LHC. The ep collisions at very low x values will unlock access to a kinematic region not explorable by other colliders with the same theoretical and experimental control, and, therefore, it constitutes a discovery frontier for novel QCD dynamics.

To illustrate the scientific potential of research in electroweak and Higgs physics at the phase-one LHeC, a few MadGraph5_aMC@NLO generator-level cross sections for unpolarised and longitudinally polarised electron beams are compared in Tab. I with those for the final LHeC. The Higgs boson case remains compelling at 0.75 TeV (especially for the charged current events), as well as the single-top-quark one, not only for mastering pioneering electron-proton measurement techniques but also for making measurements of some specific channels. For example, assuming the Run5 total integrated luminosity of 250 fb^{-1} , and an 80% electron longitudinal polarisation, one obtains the following

TABLE I. Inclusive Higgs, single top, and top pair generator level production cross sections for the phase-one and final LHeC, and for un-polarised and polarised electron beams. All cross sections are quoted at leading order, with the Higgs boson mass of 125 GeV and the top quark mass of 172 GeV.

Proton energy, E_p	7 TeV			
Electron energy, E_e	50 GeV		20 GeV	
Electron polarisation, P_e	0	-0.8	0	-0.8
$\sigma(e^-p \rightarrow H\nu_l + \text{jet} + X)$ [fb]	73	164	21	52
$\sigma(e^-p \rightarrow e^-H + \text{jet} + X)$ [fb]	14	19	3.4	5.6
$\sigma(e^-p \rightarrow e^-t\bar{t} + X)$ [fb]	26	28	0.93	0.99
$\sigma(e^-p \rightarrow \nu_l\bar{t} + X)$ [pb]	1.7	3.1	0.28	0.51

uncertainties by scaling up the errors obtained for the 1.2 TeV LHeC [4] using the numbers of produced events: $\delta\kappa_W \approx 2.5\%$, for the Higgs boson coupling to W bosons, and $\delta\mathcal{R}e(f_L^2) \approx 0.2$, for the anomalous Wtb coupling – both errors comparable to the expected final uncertainties for the corresponding measurements at the HL-LHC using 3 ab^{-1} of the proton-proton data.

Experimental conditions at the LHeC will be particularly favourable for studies of the exclusive production via photon-photon fusion; the event pile-up will be negligible, highly improving the efficiency and purity of exclusive event selection, and at the same time some sources of overwhelming backgrounds, as due to the Drell-Yan process, for example, will be absent. Moreover, no event suppression will occur due to hadronic re-scattering and the associated significant theoretical and experimental uncertainties will not intervene. Finally, data streaming (or the absence of triggering) will allow for detection and reconstruction of a much wider range of final states. The prime example is the two-photon production of W -boson pairs – as the semi-leptonic decays will be detected with high efficiency, that will lead to much improved sensitivities to anomalous couplings with respect to the measurements at the HL-LHC. In addition, excellent sensitivity is expected to the τ anomalous magnetic moment a_τ , as well as the electric dipole moment of tau-leptons.

Last but not least, already at the phase-one LHeC, also one of the key questions in nuclear physics can be addressed: how the nucleon structure is modified when immersed in a nuclear medium. For example, by a precise measurement and complete unfolding of the distributions of all parton species in a single nucleus comparable to that in the proton, the LHeC will revolutionise our understanding of nuclear structure in crucial and unique kinematic regions [1, 4]. In addition, it will allow one to clarify the dynamics for particle production in nuclear collisions in the region of relevance for the LHC.

Technological innovation – Energy Recovery Linac (ERL) technology addresses the high-power requirements to achieve high luminosity for the LHeC, i.e., recovering the energy of the electron beam of 1.2 GW power before dumping it. The fundamental principle of an ERL provides an ideal condition for a staged particle accelerator, in particular, by delaying the installation and commissioning of the 50 GeV multi-pass ERL to the final stage of the LHeC project.

Polarised electron beams can rather naturally be incorporated in an ERL type accelerator. Laser-driven state-of-the-art electron guns provide polarised beams up to 90% as previously suggested in the LHeC CDR [4]). Harmful effects like resonant depolarisation that play a major role in storage rings are by definition not existent, and the detuning of the spin precession due to the emitted synchrotron light is estimated to be on the order of a few percent. As a consequence the acceleration chain provided by the ERL is quasi spin transparent and does not play a large role in the final – longitudinal – electron beam polarisation at the IP.

Feasibility and readiness – ERL technology has matured successfully over the last half century at moderate beam power. This technology has been prioritised in the European Accelerator R&D Roadmap for particle physics, and a high-current operation of the single-pass ERL is being presently developed at the BERLinPro test facility. Construction of a 60 mA source is expected to be readily feasible for unpolarised electrons; however, such a high current source for polarised electrons needs a dedicated R&D effort.

In 2022, an extension of the ALICE experiment beyond Run4 at the HL-LHC was proposed in a letter of intent [5]. In that document, two configurations of magnetic field were considered for the new ALICE 3 detector at the IP2 – produced either by the cost effective 7.5 m long solenoid, or by the performant combination of a 2.5 m solenoid at the centre with two 2 m long dipoles on its each side. To allow for electron-hadron collisions, the dipole field is necessary for the separation of the electron beam, so the latter approach is desirable in the case that the ALICE 3

detector is used also for the electron-hadron experiment. The most recent ALICE 3 design includes a performant tracking detector that covers a wide rapidity range of $|\eta| < 4$ [6], which is well compatible with the needs for electron-hadron measurements at the stage-one LHeC. However, in contrast, the present design lacks a hadronic calorimeter (HCAL) while a performant one is obligatory for making proper measurements of the DIS charged current events and reconstruction of the decays of top quark, Higgs, W and Z bosons. Moreover, an extension of the electromagnetic calorimeter is desirable in the electron forward direction to ensure a good reconstruction of DIS neutral current events; in addition, important studies of exclusive production at the LHeC will largely benefit from specialised near-beam detectors in very forward electron and proton directions. Finally, it should be stressed that the ALICE 3 design already contains a complete data streaming mechanism (i.e. a "trigger-less" data acquisition), which is also very profitable for measurements at the LHeC. In summary, the proposed ALICE 3 detector requires relatively small modifications, apart from the addition of HCAL, to be also appropriate for the phase-one LHeC experiment.

Construction and operational costs – Following the idea of a design that can be easily extended in energy by staging additional return arcs, the basic layout of the ERL will remain the same. However, starting with one-pass ERL the return arcs 3 to 6 as well as the so-called beam spreaders and re-combiners at the end of the linacs and the bypass that guides the low-energy beam outside the particle detector will not be needed. As a result, an overall budget reduction of at least 70 MCHF is obtained for the 20 GeV version discussed here. In addition, the lower electron energy allows a much simplified design of the interaction region – in the baseline case, with $\rho \approx 1500$ m, the synchrotron radiation is relatively weak and does not require installation of sophisticated protection masks and absorbers. Last but not least, the operating costs of the 20 GeV ERL will be much smaller with respect to the 50 GeV case because of low energy losses due to synchrotron radiation and the absence of the arc 3–6 beamlines. Furthermore, the cost of the central part of the detector can be very much reduced thanks to the possibility of joining resources with the ALICE 3 project, which is also very beneficial scientifically – in addition to enabling electron-hadron studies, in effect it provides a better detector for AA research.

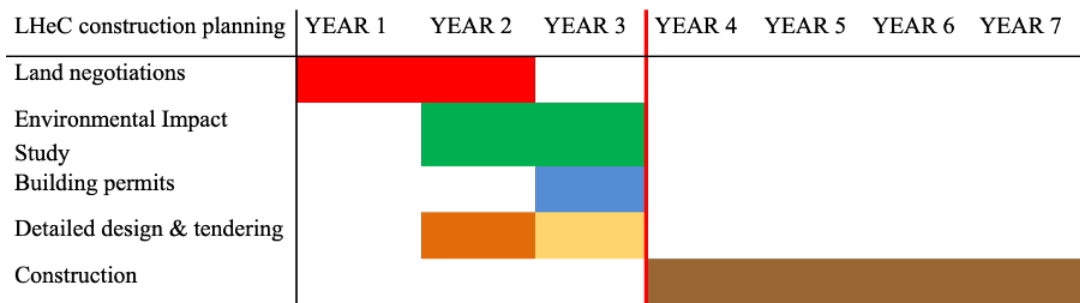


FIG. 2. Timeline for planning the construction and installation phases for the LHeC [1].

In total, 7 years are needed to build the LHeC, including permits, engineering studies, tendering and land negotiations, and the final year of connection to HL-LHC, as shown in Figure 2. Therefore, given a much simpler design of the phase-one LHeC proposed here and a timely decision on the start of such a project, we believe that it is feasible to make electron-hadron collisions available during Run5 of the HL-LHC.

Annex

The back-up document contains the article presenting the concept of phase-one LHeC as an extension of the present HL-LHC programme at CERN [2].

-
- [1] F. Ahmadova *et al.*, (2025), arXiv:2503.17727 [hep-ex].
 - [2] K. D. J. André, B. Holzer, L. Forthomme, and K. Piotrkowski, (2025), arXiv:2503.20475 [hep-ex].
 - [3] K. D. J. André *et al.*, Eur. Phys. J. C **82**, 40 (2022), arXiv:2201.02436 [hep-ex].
 - [4] P. Agostini *et al.* (LHeC, FCC-he Study Group), J. Phys. G **48**, 110501 (2021), arXiv:2007.14491 [hep-ex].
 - [5] S. Acharya *et al.* (ALICE), *Letter of intent for ALICE 3: A next-generation heavy-ion experiment at the LHC*, Tech. Rep. (CERN, Geneva, 2022) arXiv:2211.02491 [physics.ins-det].
 - [6] A. Dainese and A. Di Mauro, *Scoping document for the ALICE 3 detector*, Tech. Rep. (CERN, Geneva, 2025).